Khawar Islam

Research Scientist

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Education

2nd Sep-24 Present — **The University of Melbourne**, Victoria, Melbourne, Australia
- PhD in Computer Science and Engineering
- Advisor: Prof. Naveed Akhter (Computer Vision)
- Thesis: Benchmarking Video Understanding

2nd Mar-20 18th Feb-22 — **Sejong University**, Seoul, South Korea
- M.S in Computer Science and Engineering
- Thesis: Face Pyramid Vision Transformer
- GPA: 4.25/4.5, Presentation Link

Key Publications

CVPR 2027 (Under Review) — **DG-ViD: Benchmarking MLLMs on Video Understanding Domain Generalization**
Khawar Islam, Basim Azam, Naveed Akhtar
In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2027

ICLR 2027 (Under Review) — **CogViD: A Comprehensive Cognitive Benchmark for Evaluating Video MLLMs**
Khawar Islam, Muhammad Zaigham Zaheer, Youngmin Kim, Soyeong Jeong, Basim Azam, Muzammal Naseer, Naveed Akhtar
In International Conference on Learning Representations (ICLR), 2027

ECCV 2026 — **S^2-FracMix: Label-Preserving Self-Saliency Mixup Augmentation**
Khawar Islam, Arif Mehmood, Xin Jin, Naveed Akhter
In European Conference on Computer Vision (ECCV), 2026

ECCV 2026 — **HSFM: Hard-Set-Guided Feature-Space Meta-Learning for Robust Classification under Spurious Correlations**
Aryan Yazdan Parast, Khawar Islam, Soyoun Won, Basim Azam, Naveed Akhtar
In European Conference on Computer Vision (ECCV), 2026

ICLRw 2025 — **Context-Guided Responsible Data Augmentation with Diffusion Models**
Khawar Islam, Naveed Akhter
In International Conference on Learning Representations (ICLRw), 2025

CVPR 2024 — **DiffuseMix: Label-Preserving Data Augmentation with Diffusion Models**
Khawar Islam, Muhammad Zaigham Zaheer, Arif Mahmood, Karthik Nandakumar
In IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024

CVPR 2021 — **Image Compression with Recurrent Neural Network and Generalized Divisive Normalization**
Khawar Islam, Dang Lien Minh, Sujin Lee, Hyeonjoon Moon
In IEEE Conference on Computer Vision and Pattern Recognition Workshop (CVPRw), 2021

Research and Work Experience

1th-Ap-24 15th-Aug-26 — **Chief Technology Officer (CTO) - Computer Vision**, Upendi, Seoul, South Korea
Tools: PyTorch, OpenCV, Diffusion Models, ControlNet, IP-Adapter, FLUX, CLIP, Docker, Git
- Led independent and collaborative research in CV/ML, driving ideas through to publication.
- Designed, trained and evaluated deep-learning models for multimodal and video understanding.
- Supervises and mentors interns, junior researchers, and postgraduate-level students.
- Built and curated large multimodal datasets and benchmarks for training and evaluation.
- Participates in the preparation, authorship, and submission of research publications.
- Fine-tuned diffusion and foundation models (Stable Diffusion, FLUX, CLIP) on custom data.
- Lead team of three developers to build full-fledge product with agile methodology.
- Used vision foundation models to auto-caption data and cut manual annotation effort.
- Released code and benchmarks to support reproducible research and community use.
- Presented research results to collaborators and stakeholders, iterating from feedback.
- Data collection processes, ensuring high-quality datasets for training and validating models.

4th-Ap-22 – 30th-Ap-25 **Research Scientist - CV/ML**, Coremax Technology, Seoul, South Korea

Tools: YOLO v8, ResNet, LLMs, Chat GPT, GPT-Neo, Claude, Anime Style, CLIP, Log Analysis, IP-Adapter, Stable Diffusion, SmartEdit, FLUX, InstructPix2Pix, PhotoMaker, StoryDiffusion

- Worked on Large Language Models (GPT, NEO) for parsing Unsupervised log data.
- Developed Memory, CPU and Hardware forecasting model via Vision Transformer.
- Construct Hangul dataset (10M) and trained different OCRs to recognize handwritten samples.
- Learning Visual Features without Label/Image-Text pairs data via Self-Supervised methods.
- Described project ideas to Team Lead/CTO related to project with diagrammatic form.
- Introduced EleutherAI's/Pythia LLM to analysis DDOS attacks in unstructured data.
- Trained different ViTs, CNNs for Image Classification and Object Detection problems.
- Built UI via Gardio/Streamlit for projects to show full-fledged demo to the client.
- Managed code on internal GitHub with proper commit messages and branches for easiness.

24th-June-25 – 30th-Aug-25 **Research Associate - CV/ML**, Sejong University, Seoul, South Korea

Tools: YOLO-E, Video Depth Anything, Zero-Shot, VLMs, OWL-ViT, Panoptic DETR, Promptable Segmentation, SegFormer, Vision-Language-Action Models (VLAMs), Swin-UniNet,

- Addressed **class-imbalance**, boosting minority class 6× 200 to **1,200** (+6 × data boost)
- Improved **O.Det** performance by introducing **FLUX.1** Aug, raising mAP from 0.72 to 0.80
- Designing single quantized model Object Detection + Segmentation for delivery robot.
- Introducing adversarial perturbation concept in current Object Detection model FGSM+Fractal.
- Working on depth estimation via Depth Anything to estimate depth from footpath to road.
- Data collection processes, ensuring high-quality datasets for training and validating models.
- Plan and execute research projects, ensuring timely delivery of milestones and deliverables.
- Stay up-to-date with latest R&D in CV and DL, and apply latest techniques to solve problems.
- Implemented DINO v3 for selecting diverse scene understanding images from public dataset.

15th Ap–19 – 10th Mar-20 **Research Associate - Computer Vision**, NUCES-FAST, Karachi, Pakistan

Tools: VGG-Face, Google FaceNet, OpenFace, Facebook DeepFace, DeepID, ArcFace, Dlib.

- Developed Age-Invariant Deep Neural Network for Asian faces via OpenFace (CMU).
- Worked on Person Re-Identification/Person Search via videos captured from CCTV
- Implemented ECCV2018 for License Plate Recognition for videos in unconstrained scenarios
- Meeting with government/private institution for Data Collection and guidance.
- Data collection and preprocessing of missing children from Sarim Burney Trust International.
- Teaching and mentoring BS students to conduct research related to specific topic.
- Write clear and concise technical documentation and present solutions to stakeholders.
- Evaluate the performance of CV models and develop methodologies for model comparison.
- Lead two developers to design full-fledge demo with Web Portal and Mobile App.

1st Mar-17 – 30th Jul-18 **Senior Executive - Application Development**, Axact, Karachi, Pakistan

Tools: Swift, Xcode Editor, Alamofire, RealmSwift, SDWebImage, MessageKIT

- Designing, developing, and maintaining iOS applications using Swift programming languages.
- Writing clean, efficient, and maintainable code that follows best practices and coding standards.
- Troubleshooting and debugging issues, and implementing fixes to ensure smooth functioning.
- Participate in Agile development process, attend sprint planning, stand-ups, and retrospectives.
- Providing technical guidance and mentorship to junior team members, as needed.
- Worked on MVC and singleton pattern, real-time push notifications, Firebase, realm
- Optimizing computer vision algorithms for performance and memory usage on mobile devices.
- Maintaining and improving existing codebases, including refactoring and bug fixing.
- Developing and maintaining image processing pipelines and data processing workflows.

Honors and Achievements

2021 **NeurIPS Support**, Sponsored by Naver Labs Europe, DeepMind and Google AI

2020 **Full Tuition Fee Waiver**, School of Computer Engineering, Sejong University

2020 **Prof. Stipend**, CVPR Lab, Sejong University

2020 **Conference Travel Grant**, Ministry of Planning Commission, Pakistan

2018 **Merit Scholarship for Bachelor Degree**, Quaid-e-Azam Aligarh Society, Karachi

2016 & 2017 **Most Valuable Professional Award**, C-SharpCorner, India

2016 **Laptop Awarded, Prime Minister Pakistan Program**, Karachi, Pakistan

Summer and Winter Schools

2022 **OxML School**, Oxford Machine Learning Summer School
2022 **NYU AI School**, New York Artificial Intelligence School, USA
2021 **MLSS**, Machine Learning Summer School, Taipei, Taiwan
2021 **EEML**, Eastern European Machine Learning Summer School

English Proficiency and Relevant Courseworks

IELTS-AC 6.5 Listening 7.5, Speaking 6.5, Writing 6.5, Reading 6.0
A+ Introduction to Deep Learning, Advanced Deep Learning, Advanced AI
A0 Machine Learning, Topics in Machine Learning, Data Science
A Data Structure and Algorithms, Linear Algebra, Numerical Computing

Technical Skills

IP Algorithms Segmentation, Filter Design, Noise Removal, Compression, Super Resolution
Object Det SSD, YOLO-NAS, YOLOX, Faster R-CNN, DETR, GroundingDINO, YOLOv8
DL techniques Tracking, Optical Flow, Particle Filtering, Pose Estimation
CV Algorithms Image Classification and Detection, Tracking, Siamese Networks
Soft. Control JIRA, Agile, Scrum and GIT or Subversion
Lang Python, Keras, JAVA, Unix Shell Scripting
Deployment Docker and Kubernetes
Editor Cursor AI, Windsurf, Visual Studio Code, PyCharm, Jupyter Notebook
OS WINDOWS 10, Ubuntu 18.04, Ubuntu 20.04, MAC OSX
APIs PyTorch, TensorFlow, Caffe, OpenCV, Flask, Django

References

Dr. Zaigham Zaheer
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Prof. Muhammad Atif Tahir
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